

The Transformative Applications of ICT

A technical analysis across Education, Business, Health, and Engineering

What is ICT (Information & Communication Tech)?



Infrastructure

The hardware, networks (Internet, 5G, Wi-Fi), and data centers that form the backbone of modern computing and data transfer.



Software

The applications, operating systems, databases, and algorithms that run on the infrastructure to process information.



Data

The information that is stored, processed, and transmitted. Data is the asset enabled by ICT infrastructure and software.

Section 01: ICT in Education

From passive learning to interactive, personalized experiences.

Education: Technical Details (LMS & VR)

Learning Management System (LMS)

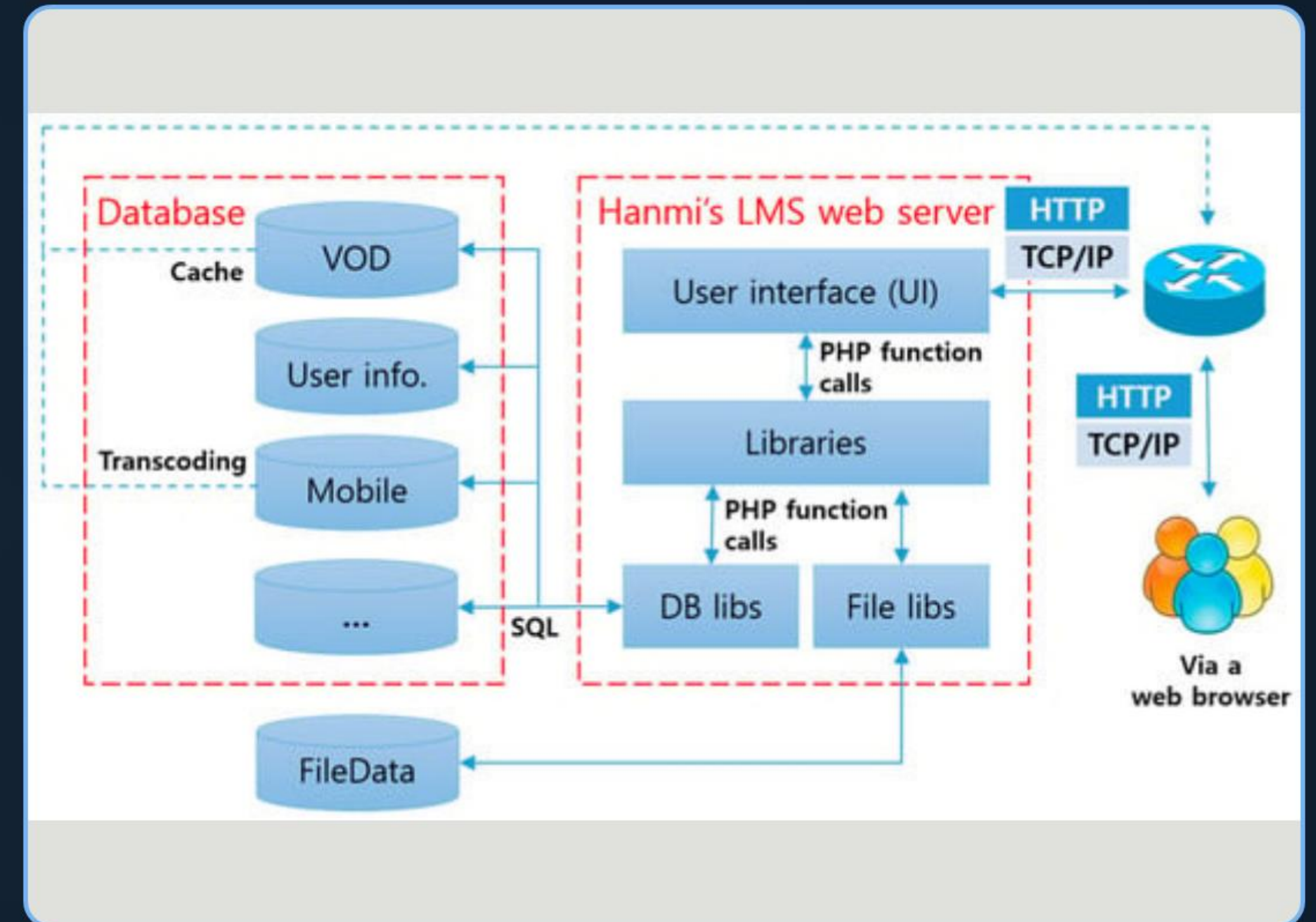
Centralizes course content, student delivery, and performance analytics.

- ✓ **Technical:** Client-server architecture with a central database (SQL/NoSQL) storing user data, grades, and content.
- ✓ **Standards:** Uses SCORM or xAPI to ensure content interoperability between different authoring tools and platforms.

Virtual Reality (VR) Labs

Immersive, safe simulations of complex environments (e.g., chemistry, surgery).

- ✓ **Technical:** Relies on 3D rendering, low-latency motion tracking, and haptic feedback controllers.
- ✓ **Analysis:** Increases retention and safety, while reducing the cost of physical labs and materials.



Section 02: ICT in Business

Driving efficiency, strategy, and global e-commerce.

Business: Technical Details (ERP & CRM)

Enterprise Resource Planning (ERP)

Integrates all core business functions (finance, HR, supply chain, manufacturing) into a single, unified system.

Analysis: Creates a 'single source of truth' for the entire organization, reducing data redundancy and automating complex workflows.

Customer Relationship Mgmt (CRM)

Manages all customer interactions, sales pipelines, and marketing data. Uses a central data warehouse to track every touchpoint.

Analysis: Leverages data analytics and AI to personalize marketing, forecast sales, and improve customer retention.

Section 03: ICT in Health (eHealth)

Enhancing diagnostics, treatment, and patient records.

Health: Technical Details (EHR & Telemedicine)

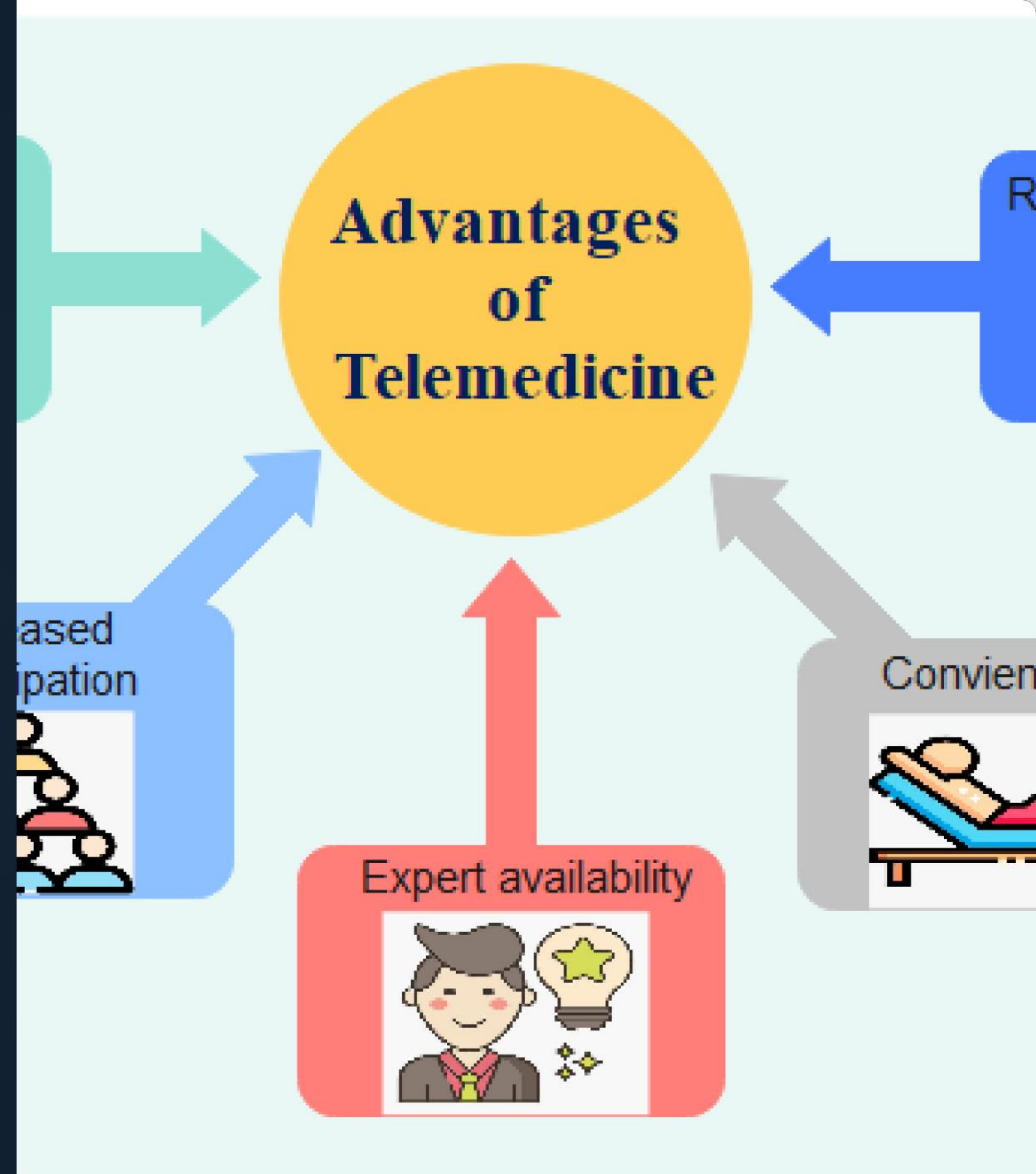
Electronic Health Records (EHR)

A digital version of a patient's complete paper chart. This is the core of all modern eHealth systems.

- » **Standards:** Data exchange is governed by standards like HL7 and FHIR, ensuring different systems can communicate.
- » **Imaging:** Radiology images (MRI, CT) are stored and shared using the DICOM standard via a PACS server.

Telemedicine

Remote clinical services using secure video conferencing and remote monitoring devices (IoT).



Section 04: ICT in Engineering

From digital design and simulation to automated manufacturing.

Engineering: Technical Details (CAD, CAM, FEA)



CAD (Design)

Computer-Aided Design. Using software to create precise 2D and 3D models of physical objects. Based on geometric modeling kernels.



CAM (Manufacturing)

Computer-Aided Mfg. Using software to control machine tools (e.g., CNC mills) based on the CAD models. Generates G-code instructions.



FEA (Analysis)

Finite Element Analysis. Simulating physical phenomena (stress, heat, flow) on a CAD model. Solves complex differential equations.

Cross-Sector Analysis: Primary Impact

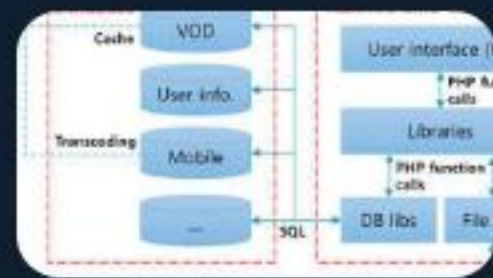


*The primary driver of ICT adoption is the ability to **process, model, and share data** securely and at scale. This leads to increased efficiency (Business), enhanced data access (Health), advanced simulation (Engineering), and personalization (Education).*

Questions?

Thank you for your attention.

Image Sources



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